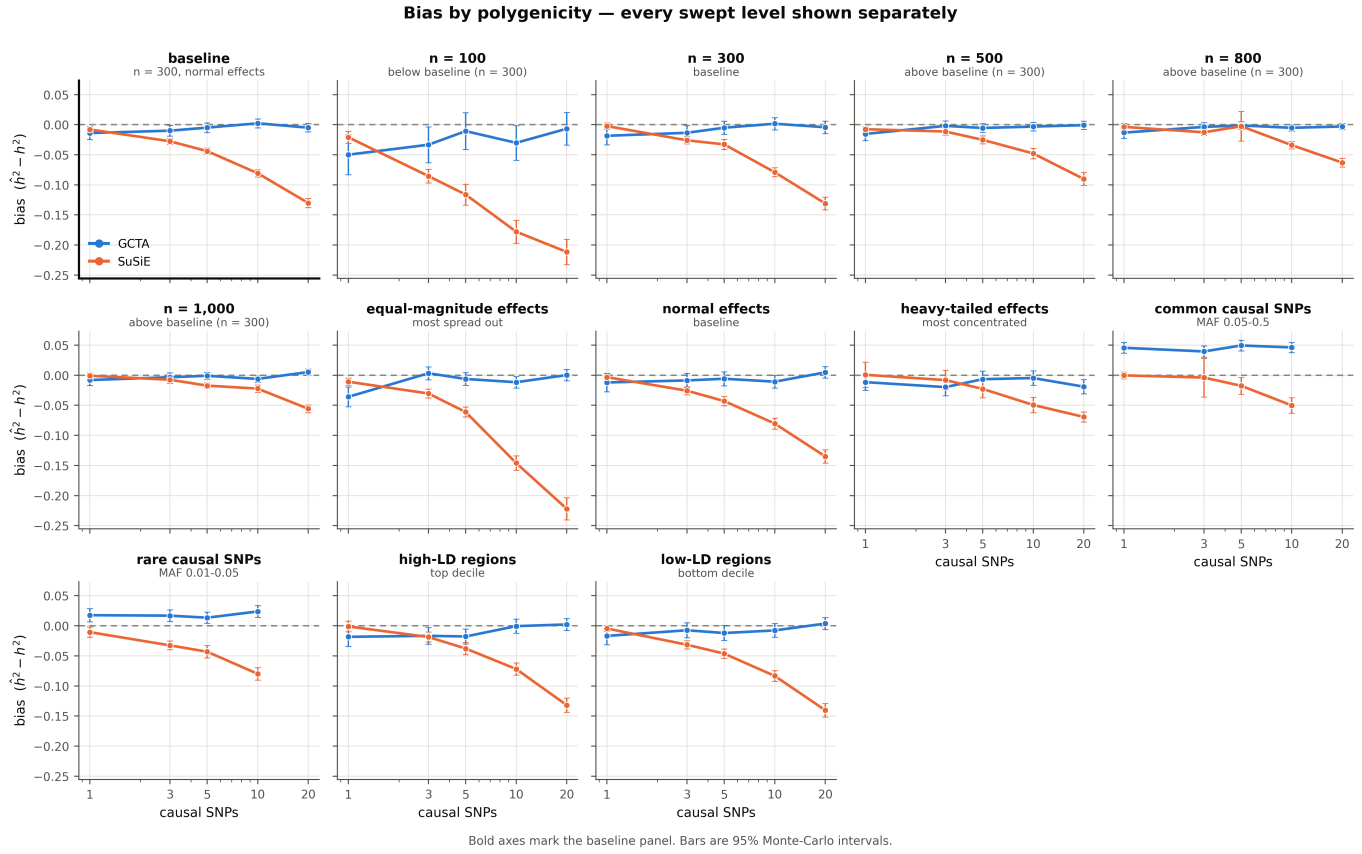
Simulation Results

Estimating the *cis*-heritability of gene expression: **GCTA/REML** (individual-level genotypes, the field benchmark) versus a **SuSiE fine-mapping** estimator (summary statistics + LD, $\hat{h}^2 = \sum(\mu_j^2 + \sigma_j^2)$).

Ground truth is not observable in real data, so everything below comes from simulation: **20,294 model fits** across **350 random 1000G EUR *cis*-regions** (±500 kb, median ~8,200 SNPs), where the true h^2 is set by construction. Both estimators see the *same* simulated phenotype at every point, so every comparison here is paired.

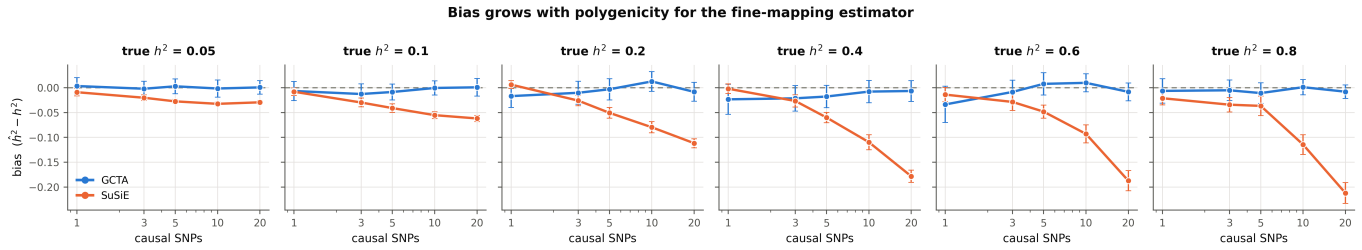
Each experimental change is reported at **every level it was swept**, with its direction stated against a baseline of *n* = 300, normally distributed effects, unrestricted causal MAF, and unselected regions.



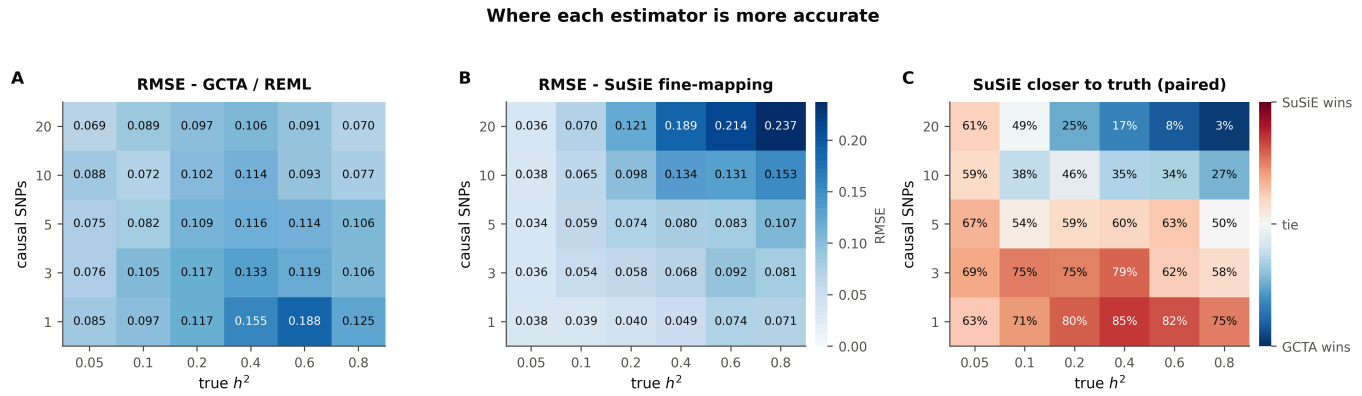
1. The switch is polygenicity

With a **single causal SNP**, fine-mapping is essentially unbiased (-0.008, vs -0.014 for GCTA) and is closer to the truth on **76% of loci**. With **20 causal SNPs** it loses 0.13 of h^2 — a 40% relative shortfall at the average true h^2 of the grid — and wins only 27% of loci. GCTA stays flat across the whole range (-0.005 at 20 causal SNPs).

The mechanism is structural, not a tuning problem. SuSiE spends a fixed budget of single-effect components on the variance it can localise; effect that is spread thinly across many variants in LD is never assigned to a credible set, and simply goes missing from $\sum(\mu^2 + \sigma^2)$. GCTA's variance component makes no attempt to localise anything, so it does not pay that penalty.



The loss is also *proportional* to h^2: at 20 causal SNPs the shortfall grows steadily with the true value, so this is a systematic under-recovery of signal, not a floor effect.



Panel C is the practical summary: fine-mapping wins the sparse, high-h^2 corner and loses the polygenic one.

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2. Sample size could repairs the bias

Sample size moves the fine-mapping bias more than any other axis tested. Going **below** baseline is punishing and going **above** it helps, monotonically:

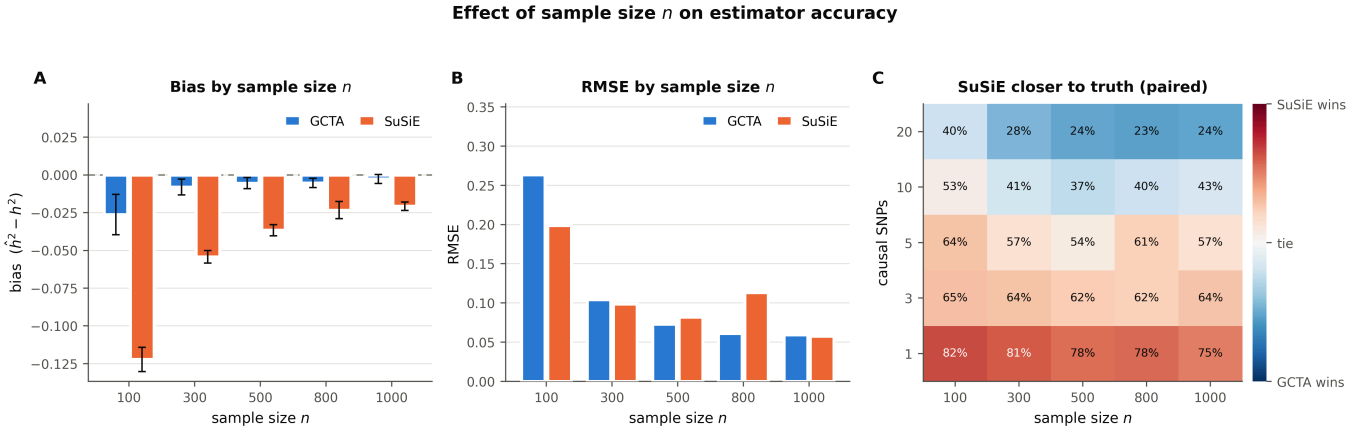
n	direction	SuSiE bias	GCTA bias	SuSiE MAE	GCTA MAE	SuSiE RMSE	GCTA RMSE	SuSiE wins
100	below baseline (n = 300)	-0.122	-0.026	0.142	0.205	0.199	0.263	61%
300	baseline	-0.054	-0.008	0.072	0.078	0.099	0.104	54%
500	above baseline (n = 300)	-0.037	-0.005	0.055	0.054	0.082	0.073	51%
800	above baseline (n = 300)	-0.023	-0.005	0.045	0.046	0.113	0.061	53%
1,000	above baseline (n = 300)	-0.021	-0.003	0.039	0.044	0.058	0.059	53%

(**MAE might be more suitable here:** SuSiE's RMSE at *n* = 800 (0.113) breaks the trend because of a *single* cell where the fit returned $\hat{h}^2$ = 4.5 (absolute error 3.7). MAE is monotone across the whole sweep — 0.142

→ 0.072 → 0.055 → 0.045 → 0.039 — and is the honest summary here.)

Fine-mapping's bias shrinks 6-fold from *n* = 100 (-0.122) to *n* = 1,000 (-0.021): with more samples the marginal signal is sharp enough for SuSiE to resolve, so less effect goes unassigned.

But **bias and usefulness move in opposite directions.** At *n* = 100 — where SuSiE's bias is at its worst — its error is 31% *smaller* than GCTA's (MAE 0.142 vs 0.205) and it wins 61% of loci, because REML is very noisy in small samples. By *n* = 1,000 the bias is nearly gone, but so is the advantage: only 11% better on MAE (0.039 vs 0.044) and 53% of loci. **Fine-mapping is most worth using exactly where it is most biased.** It trades REML's variance for its own bias, and that trade pays only while REML is noisy.



3. Effect concentration

A clarification, because the axis name invites the wrong reading: the three effect-size priors do **not** carry different amounts of genetic effect. Causal effects are rescaled so that $\beta' R \beta = h^2$ exactly in every arm, so total effect is identical by construction. What changes is how that fixed total is **spread across the causal SNPs**:

prior	how the effect is distributed
equal-magnitude (`fixed`)	every causal SNP identical — maximally spread
normal (`normal`)	varied magnitudes — the baseline
heavy-tailed (`pareto`)	one or two dominate — maximally concentrated

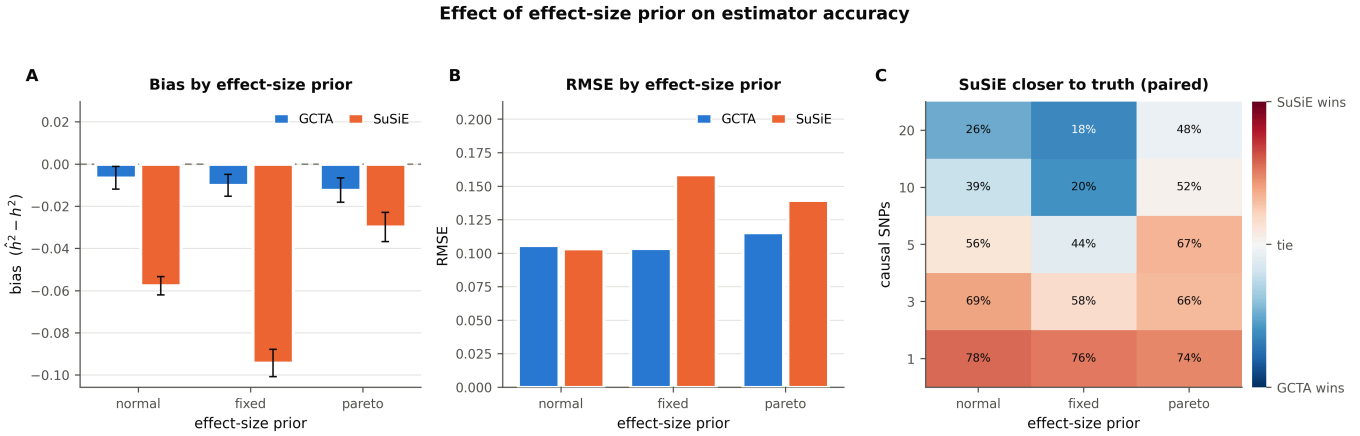
(Effect *magnitude* is already covered twice elsewhere: total magnitude is the h^2 axis, and per-causal-SNP magnitude is $\sqrt{h^2/k}$, i.e. the causal-count axis.)

prior	concentration	SuSiE bias	SuSiE bias at 20 causal	SuSiE wins
equal-magnitude effects	most spread out	-0.094	-0.222	43%
normal effects	baseline	-0.058	-0.135	53%
heavy-tailed effects	most concentrated	-0.030	-0.069	61%

This is the same mechanism as §1, seen along a second axis. Concentration is what fine-mapping can exploit: with heavy-tailed effects it recovers most of the signal (-0.030) and wins 61% of loci; with equal-

magnitude effects spread evenly it is at its worst (-0.094) and drops to 43% — the only condition tested where GCTA is preferable on average. GCTA is indifferent to the prior (-0.010 vs -0.012), as expected.

Practically: **20 causal SNPs of equal size is a near-worst case** (-0.222), while 20 causal SNPs dominated by one or two is much more benign (-0.069). The count of causal variants alone does not predict the error — how the effect is distributed among them matters just as much.



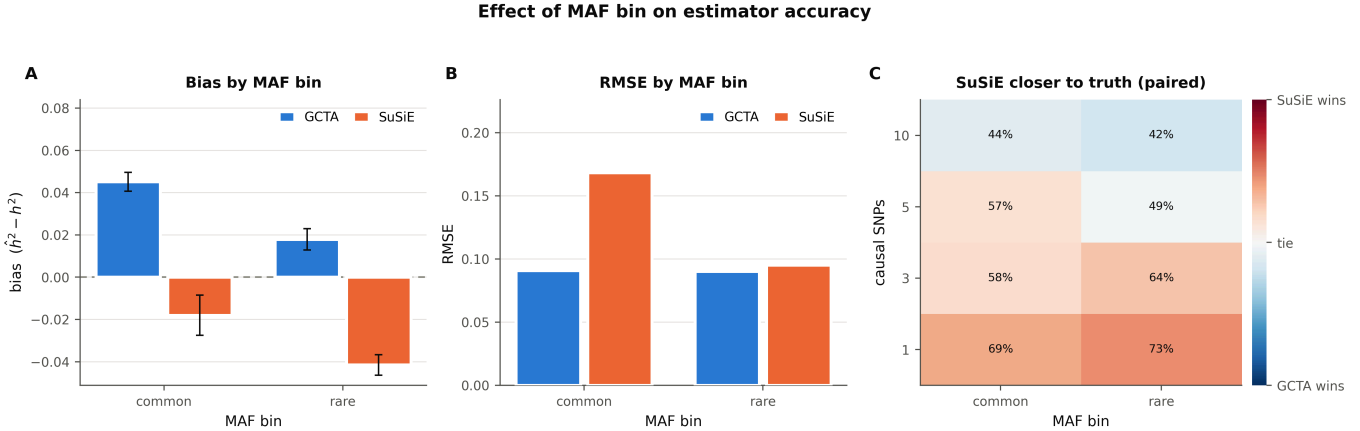
4. Causal allele frequency

Every other condition leaves GCTA near-unbiased. Restricting causal variants to a frequency bin does not:

causal SNPs drawn from	GCTA bias	SuSiE bias	SuSiE wins
common causal SNPs (MAF 0.05-0.5)	**+0.045**	-0.018	57%
rare causal SNPs (MAF 0.01-0.05)	**+0.018**	-0.042	57%

GCTA turns **upward**-biased (+0.045 for common causal variants) — the only sign flip anywhere in the study. The GRM here weights all variants equally, which implicitly assumes effect size is unrelated to allele frequency; when causal variants are confined to one frequency bin that assumption is false and the variance component is mis-attributed. This is the known GCTA MAF/LD-stratification issue, and it is the argument for MAF-stratified GRMs (GREML-LDMS) rather than a single GRM.

Worth flagging because it is invisible in the other campaigns: the benchmark itself is not unconditionally trustworthy, so "SuSiE differs from GCTA on real data" cannot be read as "SuSiE is wrong" without knowing the frequency architecture.



5. LD architecture

regions	SuSiE bias	GCTA bias	SuSiE wins
low-LD regions (bottom decile)	-0.061	-0.008	54%
high-LD regions (top decile)	-0.053	-0.010	54%
unselected (baseline)	-0.058	-0.006	54%

Top- versus bottom-decile LD separates the fine-mapping bias by only 0.009 — an order of magnitude less than the causal-count axis, and less than moving *n* from 300 to 500. LD is the thing fine-mapping is built to resolve, and on this evidence it resolves it: what defeats the estimator is signal being *diffuse*, not signal being *correlated*.

One caveat on the size of this contrast. The LD score used here is the fraction of sampled SNP pairs with $r^2 \geq 0.8$, and across 300 random ±500 kb windows it is tightly distributed (median 0.0024, p90 0.0051, max 0.0109). Top and bottom deciles of a narrow distribution are not a dramatic contrast; a stronger test would select on LD-block structure directly rather than on this scalar.

6. The two estimators fail in opposite ways

A heritability outside $[0, 1]$ is not a heritability. Both estimators produce them, at very different rates and severities:

condition	GCTA out of range	SuSiE out of range	worst SuSiE error
baseline	7.9%	0.2%	0.48
n = 100	22.2%	0.1%	1.46
n = 300	7.2%	0.2%	0.48
n = 500	4.0%	0.2%	0.97
n = 800	2.2%	0.3%	3.69
n = 1,000	0.9%	0.2%	0.42

GCTA fails often but mildly. With `--reml-no-constrain`, REML is free to return a negative variance component, and at *n* = 100 it does so on 22% of fits. These are small negatives around a true h^2 near zero — statistically well-behaved, and the standard reason not to truncate them before averaging.

SuSiE fails rarely but catastrophically. It goes out of range on well under 1% of fits, but when it does the estimate can be 3–4× larger than any possible heritability — while reporting clean convergence and a credible set with PIP ≈ 1. This is the rank-deficient panel LD (~8,200 SNPs, 503 reference individuals) letting the IBSS fit run away. It is the same failure mode that `susie_rss`'s `check_prior` guard catches in its severe form, in a milder version that slips past.

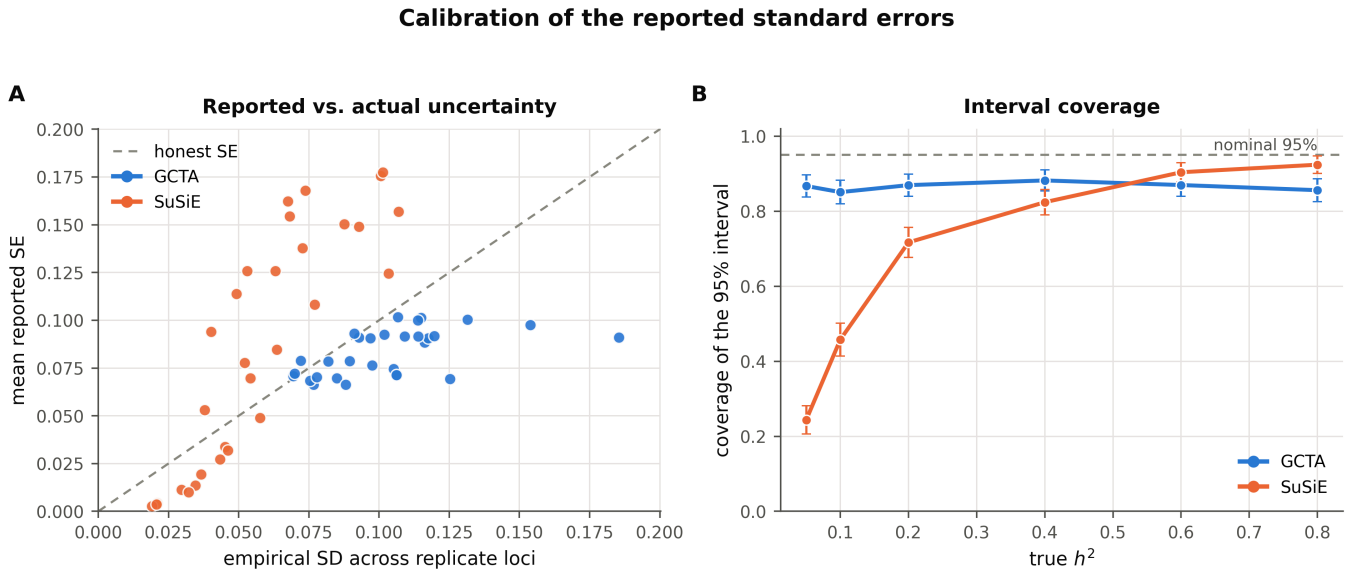
Consequences worth acting on: **report MAE alongside RMSE** (a handful of runaway fits dominate a squared-error summary), and **screen fine-mapping h^2 estimates against $[0, 1]$** in any production run rather

than trusting the convergence flag. `tables/` and `conditions.csv` carry `frac_impossible` and `max_abs_err` for every condition.

Interval coverage

Nominal 95% intervals cover the truth 87% of the time for GCTA and 68% for SuSiE. They fail for opposite reasons:

- **GCTA** reports standard errors that are honest in shape but slightly too small relative to the spread actually seen across replicate loci.
- **SuSiE**'s analytic SE, $\sqrt{\sum \mathrm{Var}(\beta_j^2)}$ under a normal posterior, is *larger* than the empirical spread at low h^2 — yet coverage there is worst, because the interval is centred on a badly biased point estimate. A wide interval around the wrong number still misses.



The practical consequence: the SuSiE SE should not be used for inference or meta-analysis weighting as it stands. It quantifies posterior uncertainty about the effects it found, not uncertainty about the effects it missed — and the effects it missed are the whole problem.

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7. What this implies for real data

situation	use	why
Small eQTL study ($n \lesssim 300$)	**SuSiE**, expect downward bias	REML variance dominates; SuSiE has ~30% lower MAE despite the bias
Large study ($n \gtrsim 800$)	**either**, GCTA if unbiasedness matters	the two converge; SuSiE keeps a small residual downward bias
Sparse architecture expected	**SuSiE**	76% win rate at one causal variant
Polygenic *cis* architecture	**GCTA**	SuSiE loses ~0.13 of h^2 at 20 causal SNPs

situation	use	why
Frequency-stratified architecture	**neither, unmodified**	GCTA turns upward-biased; use a MAF-stratified GRM
Any reported h^2 standard error	**do not trust as-is**	coverage is 68–87%, not 95%
Any fine-mapping h^2 in production	**screen against [0, 1]**	rare runaway fits report clean convergence

The general point: on real data the two estimators disagree most for polygenic, low-heritability genes — and that disagreement is *predictable* in size and direction, so it can be corrected for rather than treated as noise.

Something worth noticing

SuSiE used about 15% fewer SNPs than GCTA. The old quality-control filter removes A/T and C/G SNPs because their strand direction can be ambiguous. However, the summary statistics and reference data come from the same .bim file, so this filter is unnecessary here and may slightly disadvantage SuSiE.

The simulated genotypes are normally distributed rather than real genetic data. They preserve the LD pattern, but not the real minor allele frequency or haplotype structure. We plan to check this using GTEx data, but that analysis has not yet been moved to the new pipeline.

Both of the first two are one-line config changes; each costs about 25 minutes of compute.
